

BS-CH101 / BS-CH201 — Chemistry-I

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Module 3 Long Notes: Intermolecular Forces, Real Gases and Critical Phenomena

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1. Intermolecular Forces: Classification and Origin

Intermolecular forces (IMF) are attractive or repulsive forces between molecules, atoms, or ions that are not covalently bonded. They arise from electrostatic interactions between charge distributions and are the cause of condensation, surface tension, viscosity, solubility, and deviation from ideal-gas behaviour. They are *weaker* than intramolecular (covalent/ionic) bonds but collectively determine the physical state and material properties of matter.

1.1 Classification of Intermolecular Forces

All intermolecular forces can be traced to Coulomb's law: $V \propto q_1 q_2 / r$. Depending on whether the charges are permanent, induced, or fluctuating, we get different categories.

Force	Between	Origin	Example
Ion-ion	Two ions	Full charges $q_1 q_2$	NaCl lattice
Ion-dipole	Ion + polar molecule	Charge + permanent dipole	Na^+ –water hydration
Dipole-dipole (Keesom)	Two polar molecules	Two permanent dipoles	HCl–HCl, acetone
Dipole-induced dipole (Debye)	Polar + polarisable molecule	Permanent dipole induces temporary dipole	HCl–Ar
Dispersion (London)	Any two molecules	Instantaneous dipole-induced dipole	Ar–Ar, CH_4
Hydrogen bonding	H covalently bonded to F, O, N	Electrostatic + partial covalent	Water, HF, DNA

1.2 Ionic Interactions

An ionic interaction is the electrostatic attraction between oppositely charged ions. It is the strongest non-covalent force.

Coulomb Potential Between Two Ions

$$V_{\text{ion-ion}} = -\frac{z_1 z_2 e^2}{4\pi\epsilon_0 r}$$

z_1, z_2 = signed charge numbers; $e = 1.602 \times 10^{-19}$ C; r = interionic distance; $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$.

Energy falls as $1/r$ (long range). In a crystal, many-ion Madelung sums give the lattice energy.

Lattice energy and Born–Haber cycle: The Born–Haber cycle uses Hess's law to evaluate

the lattice energy U_L indirectly, since direct measurement is impossible:

$$U_L = \Delta H_f^\circ(\text{salt}) - \Delta H_{\text{sub}} - \Delta H_{\text{IE}} - \frac{1}{2}\Delta H_{\text{diss}} - \Delta H_{\text{EA}}$$

Higher lattice energy means harder crystal, higher melting point. MgO ($U_L \approx 3850 \text{ kJ mol}^{-1}$) is harder than NaCl ($U_L \approx 787 \text{ kJ mol}^{-1}$) because $\text{Mg}^{2+}/\text{O}^{2-}$ charges are double.

1.3 Dipole Moment and Polar Molecules

Dipole moment $\vec{\mu}$ arises whenever positive and negative charge centres do not coincide. For a diatomic A–B: $\mu = q \cdot d$, where q is the partial charge and d is the bond length. Units: debye (D), $1 \text{ D} = 3.336 \times 10^{-30} \text{ C m}$.

Molecule	μ (D)	Comment
H_2O	1.85	Bent structure, dipoles add
NH_3	1.47	Pyramidal
HCl	1.08	Heteronuclear diatomic
CO_2	0	Linear, dipoles cancel
CCl_4	0	Tetrahedral, dipoles cancel
CHCl_3	1.01	Asymmetric tetrahedral

Why CO_2 and CCl_4 are non-polar: Each individual bond is polar but the vector sum of all bond dipoles is zero due to the symmetric geometry.

1.4 Dipole–Dipole (Keesom) Interaction

Two polar molecules attract when oppositely charged ends face each other and repel when like ends face. After thermal averaging over all orientations:

Keesom Orientational Energy (Thermal Average)

$$V_{\text{Keesom}} = -\frac{2\mu_1^2\mu_2^2}{3(4\pi\epsilon_0)^2k_B T r^6}$$

Key feature: falls as r^{-6} and is temperature-dependent (decreases at higher T because thermal motion disrupts orientation).

1.5 Dipole–Induced Dipole (Debye) Interaction

A polar molecule distorts the electron cloud of a nearby polarisable, non-polar molecule creating an induced dipole. The interaction is always attractive.

Debye Induction Energy

$$V_{\text{Debye}} = -\frac{\mu^2\alpha'}{(4\pi\epsilon_0)^2 r^6}$$

α' = polarisability volume (SI: $\text{C}^2 \text{m}^2 \text{J}^{-1}$; often tabulated in $\text{\AA}^3$). Like Keesom, falls as r^{-6} but is *temperature-independent*.

1.6 London Dispersion Forces

Even completely non-polar atoms/molecules (noble gases, hydrocarbons) attract each other and can be liquefied. This requires a quantum-mechanical explanation.

London dispersion (induced-dipole–induced-dipole) force: At any instant, the electron distribution in an atom fluctuates, creating a transient dipole. This instantaneous dipole polarises a neighbouring atom, inducing a correlated dipole. The time-averaged interaction is always *attractive*.

London Dispersion Energy

$$V_{\text{London}} = -\frac{3}{2} \frac{\alpha'_1 \alpha'_2}{(4\pi\epsilon_0)^2 r^6} \cdot \frac{I_1 I_2}{I_1 + I_2}$$

I_1, I_2 = ionisation energies; α'_1, α'_2 = polarisability volumes.

Key: falls as r^{-6} ; increases with polarisability (larger molecules, more electrons, heavier atoms).

Why dispersion forces increase down a group:

Noble gas	Polarisability (pm^3)	Boiling point (K)	Trend
He	2.3	4.2	
Ne	4.0	27.1	
Ar	16.3	87.3	Increasing down
Kr	24.8	119.9	
Xe	40.4	165.1	

Solved Example: Why Hexane Boils at 69°C but Pentane at 36°C

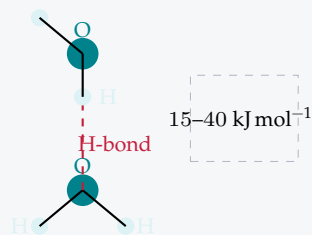
Both are non-polar and rely on London dispersion only. Hexane (C_6H_{14} , MW = 86) has more electrons and higher polarisability than pentane (C_5H_{12} , MW = 72). The stronger London forces require more energy to overcome, raising the boiling point by $\sim 33^\circ\text{C}$.

1.7 Hydrogen Bonding

Hydrogen bonding is the strongest intermolecular interaction between neutral molecules ($15\text{--}40 \text{ kJ mol}^{-1}$, about 10–20% of a covalent bond). It requires:

1. A hydrogen covalently bonded to a highly electronegative atom (F, O, or N) — the *donor*.
2. A nearby lone pair on another electronegative atom — the *acceptor*.

Hydrogen Bonding in Water



Consequences of hydrogen bonding:

- Anomalous high boiling points: water (100°C) vs H_2S (-60°C), HF (19.5°C) vs HCl (-85°C).
- High heat of vaporisation of water (44 kJ mol^{-1}).
- Ice is less dense than liquid water (open hydrogen-bonded network).
- Protein secondary structure (alpha helix: i to $i + 4 \text{ N-H}\cdots\text{O}=\text{C}$).
- DNA double helix held by A–T (2 H-bonds) and G–C (3 H-bonds) base pairing.

Intermolecular vs intramolecular H-bonding:

Type	Occurs between	Effect
Intermolecular	Different molecules	Higher bp, greater association in solution
Intramolecular	Within same molecule	Lower bp than without H-bond; less association

Example: *ortho*-nitrophenol has an intramolecular H-bond between OH and the adjacent NO_2 ; *para*-nitrophenol cannot, so it forms intermolecular H-bonds and has a higher melting point.

1.8 Polarisability and Its Determinants

Polarisability α measures how easily the electron cloud of an atom or molecule is distorted by an external electric field. Induced dipole $\mu_{\text{ind}} = \alpha E$.

Polarisability increases with:

- Increasing number of electrons (larger atoms, heavier atoms down a group).
- Larger atomic radius.
- π -electron systems (benzene, graphene).
- Ease of electron delocalisation.

1.9 Relative Strengths: Summary Table

Force	Range	Energy (kJ mol^{-1})	r dependence
Ion-ion	Long	250–1000	r^{-1}
Ion-dipole	Medium	15–85	r^{-2}
H-bond	Short-medium	15–40	r^{-2} – r^{-3}
Dipole-dipole	Short	3–20	r^{-6} (avg.)
Dipole-induced	Short	1–5	r^{-6}
London dispersion	Short	0.1–10	r^{-6}

Exam Points (Section 1)

- Arrange forces: London < dipole-dipole < H-bond < ion-dipole < ion-ion.
- Keesom and London both vary as r^{-6} ; Keesom is temperature-dependent.
- Polarisability increases down a group; explains increasing bp of noble gases.
- H-bond needs F, O, or N on both donor and acceptor.
- $\mu = qd$ in debye ($1 \text{ D} = 3.336 \times 10^{-30} \text{ C m}$); zero for symmetric molecules.
- Born-Haber cycle: use it to evaluate lattice energy of ionic solids.

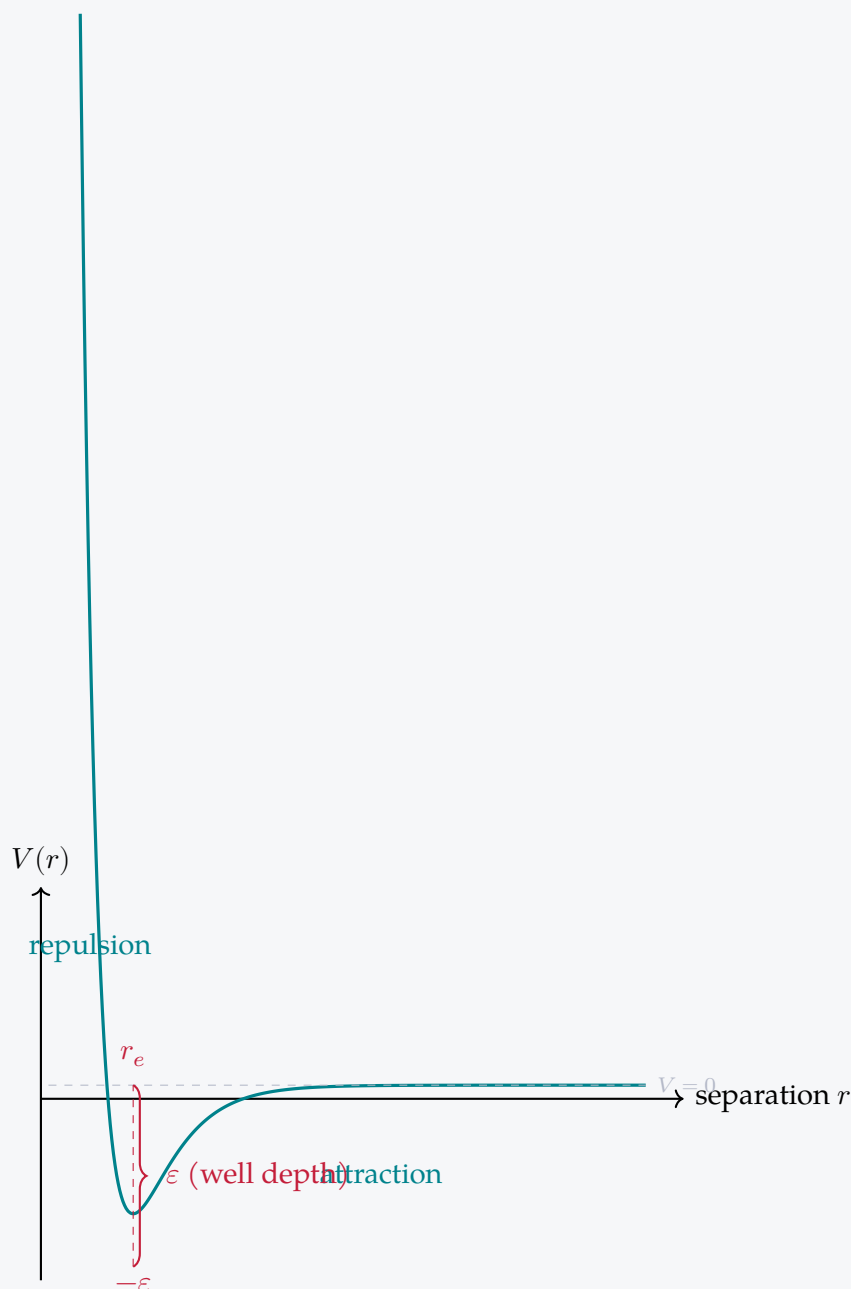
2. Potential Energy Curves and the Lennard-Jones Model

A **potential energy curve** $V(r)$ describes how the potential energy of a pair of molecules (or atoms) changes with their separation distance r . It is the single most important concept linking molecular theory to macroscopic properties.

2.1 Qualitative Shape of the Potential Energy Curve

At very large r : $V(r) \rightarrow 0$ (no interaction). As r decreases, London/dipole attraction lowers V . At short range, electron-cloud repulsion rises steeply. The curve has a minimum at the equilibrium separation r_e .

Potential Energy Curve: Key Features



2.2 Force from the Potential Energy Curve

Force and Potential Energy

$$F(r) = -\frac{dV}{dr}$$

At $r = r_e$: $F = 0$, V is minimum (equilibrium).

At $r < r_e$: $dV/dr < 0$, so $F > 0$ (repulsive).

At $r > r_e$: $dV/dr > 0$, so $F < 0$ (attractive).

Solved Example: Equilibrium Separation

A model potential has $V(r) = \varepsilon [(r_0/r)^{12} - 2(r_0/r)^6]$. Find the equilibrium separation.

Solution: Set $dV/dr = 0$:

$$\frac{dV}{dr} = \varepsilon \left[-12 \frac{r_0^{12}}{r^{13}} + 12 \frac{r_0^6}{r^7} \right] = 0 \quad \Rightarrow \quad \frac{r_0^{12}}{r^{13}} = \frac{r_0^6}{r^7} \quad \Rightarrow \quad r_e = r_0$$

The equilibrium separation equals the parameter r_0 .

2.3 The Lennard-Jones 12-6 Potential

The most widely used pair potential in chemistry and molecular simulation is:

Lennard-Jones (12-6) Potential

$$V_{\text{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

- ε = well depth (energy unit), measure of attraction strength.
- σ = the separation at which $V_{\text{LJ}} = 0$ (effective diameter).
- $r_e = 2^{1/6}\sigma \approx 1.122\sigma$ (equilibrium separation).
- $(\sigma/r)^{12}$ = short-range repulsion (Pauli exclusion / electron-cloud overlap).
- $(\sigma/r)^6$ = long-range London dispersion attraction.

Pair	ε/k_B (K)	σ (pm)
Ar–Ar	120	341
Kr–Kr	171	360
N ₂ –N ₂	95	370
CH ₄ –CH ₄	148	382

Limitations: The LJ potential ignores many-body effects, anisotropy, and electrostatics. For polar molecules or water, more elaborate potentials (TIP4P, SPC/E) are needed in simulations.

2.4 Morse Potential for Chemical Bonds

For vibrating diatomic molecules the Morse potential gives a more realistic description than the harmonic oscillator (seen in Module 2):

Morse Potential

$$V_{\text{Morse}}(r) = D_e [1 - e^{-\beta(r-r_e)}]^2$$

D_e = dissociation energy measured from the bottom of the well; $\beta = \omega_e \sqrt{\mu/2D_e}$ relates to the harmonic vibrational frequency ω_e .

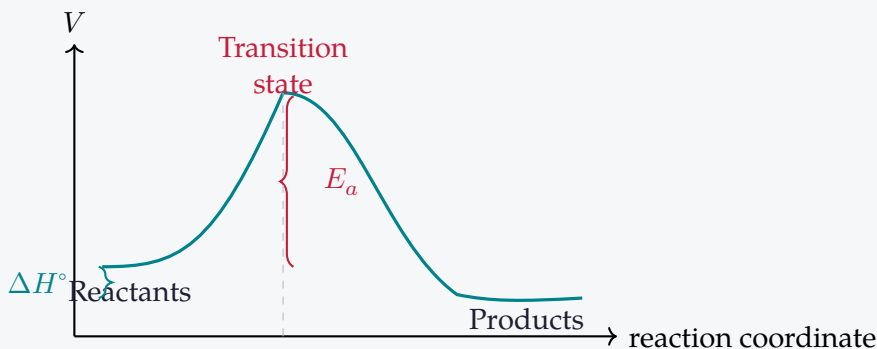
Advantage over harmonic: reproduces dissociation, anharmonicity, and unequal level

spacing.

2.5 Potential Energy Surface (PES) for Chemical Reactions

For a reaction like $A + BC \rightarrow AB + C$, the energy depends on two distances r_{AB} and r_{BC} — forming a 2D surface in 3D space.

Reaction Path on a Potential Energy Surface



Key features of a PES:

- **Minima** = stable molecules (reactants, products, intermediates).
- **Saddle points** = transition states (highest energy along the reaction path).
- **Activation energy** E_a = energy barrier from reactants to the saddle point.
- **Reaction enthalpy** ΔH° = difference in energy between reactants and products.
- An *exothermic* reaction has products at lower energy; *endothermic* has products higher.

2.6 Transition State Theory (Activated Complex Theory)

Transition state theory (TST, Eyring, 1935) treats the saddle-point configuration as a quasi-equilibrium “activated complex” $[ABC]^\ddagger$.

Eyring Rate Equation

$$k = \frac{k_B T}{h} e^{-\Delta G^\ddagger / RT} = \frac{k_B T}{h} e^{\Delta S^\ddagger / R} e^{-\Delta H^\ddagger / RT}$$

$\Delta G^\ddagger$, $\Delta H^\ddagger$, $\Delta S^\ddagger$ are the free energy, enthalpy, and entropy of activation.
The factor $k_B T / h \approx 6.3 \times 10^{12} \text{ s}^{-1}$ at 298 K is the attempt frequency.

Connection to Arrhenius: Comparing with $k = A e^{-E_a / RT}$ gives $E_a \approx \Delta H^\ddagger + RT$ and the pre-exponential factor $A = (k_B T / h) e^{1 + \Delta S^\ddagger / R}$.

Exam Points (Section 2)

- Draw the LJ potential; label σ , ε , $r_e = 2^{1/6} \sigma$.
- At r_e : V minimum, $F = 0$. For $r < r_e$: repulsion. For $r > r_e$: attraction.

Frequently Asked / Viva Questions (contd.)

- Morse potential adds anharmonicity and dissociation — improves on harmonic oscillator.
- On a PES: minima = stable species, saddle points = transition states.
- Eyring equation: $k = (k_B T/h) \exp(-\Delta G^\ddagger/RT)$.

3. Equations of State for Real Gases

An **equation of state** (EOS) is a mathematical relationship between the state variables P , V , T , and n of a substance. The ideal gas law ($PV = nRT$) is the simplest EOS but fails at high pressure and low temperature. *Real-gas* equations of state introduce correction terms that account for finite molecular size and intermolecular attractions.

3.1 The Ideal Gas Law and Its Assumptions

The ideal gas law rests on two assumptions that break down at real conditions:

1. Molecules are point particles (zero volume).
2. No intermolecular forces (no attraction or repulsion).

Ideal Gas Law

$$PV = nRT, \quad R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$$

3.2 Van der Waals Equation of State

Van der Waals (1873) introduced two correction parameters:

- b (**co-volume**): excluded volume per mole due to finite molecular size. Corrects V .
- a (**attraction parameter**): pressure is reduced by attraction, so an internal pressure term an^2/V^2 is added. Corrects P .

Van der Waals Equation

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

For one mole:

$$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

$V_m = V/n$ is the molar volume; a in $\text{L}^2 \text{ atm mol}^{-2}$; b in L mol^{-1} .

Physical interpretation:

- $(V_m - b)$: the free volume available to molecules is the total volume minus what the molecules themselves occupy.
- a/V_m^2 : molecules near the wall are pulled back by neighbours — the measured pressure P is less than the kinetic-theory pressure P_{kin} , so we add back a/V_m^2 .

Gas	a ($\text{L}^2 \text{ atm mol}^{-2}$)	b (L mol^{-1})
H_2	0.244	0.0266
N_2	1.39	0.0391
CO_2	3.59	0.0427
H_2O	5.46	0.0305
He	0.034	0.0237

Solved Example: Pressure of Real vs Ideal Gas

Problem: 1 mol of CO_2 in a $V = 1.0$ L container at $T = 500$ K. Compare ideal and van der Waals pressures.

Ideal:

$$P_{\text{ideal}} = \frac{RT}{V_m} = \frac{0.08206 \times 500}{1.0} = 41.0 \text{ atm}$$

Van der Waals ($a = 3.59$, $b = 0.0427$):

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2} = \frac{0.08206 \times 500}{1.0 - 0.0427} - \frac{3.59}{1.0^2} = 42.8 - 3.6 = \mathbf{39.2 \text{ atm}}$$

The real gas pressure is lower because the net effect of attractions (a) exceeds the volume correction (b) at this condition.

3.3 Compressibility Factor Z

Compressibility Factor

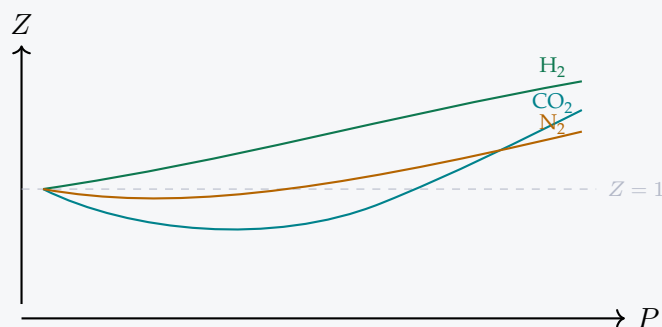
$$Z = \frac{PV_m}{RT} = \frac{PV}{nRT}$$

Ideal gas: $Z = 1$ at all P .

Real gas: $Z < 1$ when attractions dominate (usually at low–moderate P).

Real gas: $Z > 1$ when finite volume (repulsion) dominates (very high P).

Compressibility Factor vs Pressure for Real Gases



3.4 Virial Equation of State

The virial equation expresses Z as a power series in density $1/V_m$:

Virial Equation of State

$$Z = \frac{PV_m}{RT} = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

B = second virial coefficient (accounts for pair interactions), C = third virial coefficient.
 For the van der Waals gas at low density: $B = b - a/(RT)$.
 At the Boyle temperature T_B : $B = 0$ so $Z \approx 1$ at low-moderate pressure.

3.5 Other Equations of State

EOS	Expression (1 mol)	Improvement over VdW
Dieterici	$P(V_m - b) = RT e^{-a/RTV_m}$	Better near critical point
Redlich-Kwong	$P = \frac{RT}{V_m - b} - \frac{a}{T^{1/2}V_m(V_m + b)}$	Better for gas-phase calculations
Peng-Robinson	Modified cubic with two parameters	Used in petroleum engineering
Ideal gas	$PV_m = RT$	Reference baseline

3.6 Boyle Temperature

At the Boyle temperature, the second virial coefficient $B = 0$, so the gas behaves nearly ideally over a range of pressures.

Boyle Temperature for a van der Waals Gas

$$T_B = \frac{a}{Rb}$$

Solved Example: Boyle Temperature of CO₂

$$a = 3.59 \text{ L}^2 \text{ atm mol}^{-2}, b = 0.0427 \text{ L mol}^{-1}:$$

$$T_B = \frac{3.59}{0.08206 \times 0.0427} = \frac{3.59}{0.003504} = 1024 \text{ K}$$

Below T_B (1024 K for CO₂), attractions dominate and $Z < 1$ at modest pressures.

3.7 Joule–Thomson Effect

When a real gas expands irreversibly through a porous plug (throttle) from high pressure to low, the process is isenthalpic ($\Delta H = 0$). The temperature may rise or fall.

Joule–Thomson Coefficient

$$\mu_{JT} = \left(\frac{\partial T}{\partial P} \right)_H$$

For a van der Waals gas at low density:

$$\mu_{JT} \approx \frac{1}{C_P} \left(\frac{2a}{RT} - b \right)$$

At $T < T_i$ (inversion temperature): $\mu_{JT} > 0$, gas cools on expansion.

At $T > T_i$: $\mu_{JT} < 0$, gas warms on expansion.

Condition	μ_{JT}	Effect
$T < T_i$	Positive	Cooling (used in liquefaction)
$T = T_i$	Zero	No temperature change
$T > T_i$	Negative	Heating

Inversion temperature: $T_i = 2a/(Rb)$. Note $T_i = 2T_B$. For N₂: $T_i \approx 621 \text{ K}$, so pre-cooling is needed before throttling to liquefy nitrogen. For H₂ and He: T_i is below room temperature, so they warm on throttling without pre-cooling.

Linde Liquefaction Process and Joule–Thomson Effect

The Linde process uses the Joule–Thomson effect to liquefy air: compressed gas expands through a throttle valve, cools, is recycled back, and after many cycles reaches the liquefaction temperature. Gas must be pre-cooled below its inversion temperature first. This principle is used to produce liquid N₂ (77 K), liquid O₂ (90 K), and liquid air for cryogenics.

Exam Points (Section 3)

- Van der Waals: a corrects P (attractions), b corrects V (molecular size).
- Solve $P = RT/(V_m - b) - a/V_m^2$ numerically in 5-mark problems.

Frequently Asked / Viva Questions (contd.)

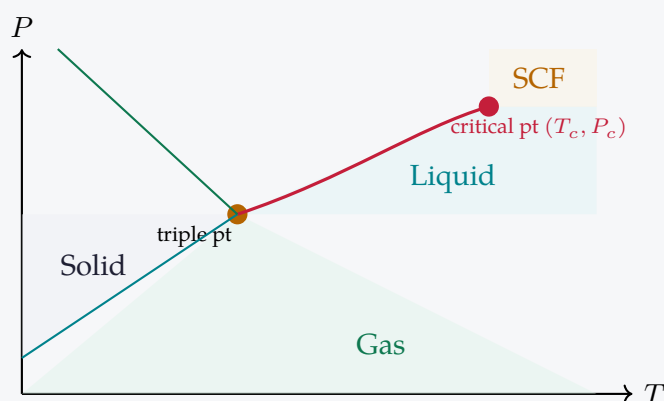
- $Z < 1$: attraction-dominated; $Z > 1$: repulsion-dominated.
- Boyle temperature: $T_B = a/(Rb)$; $B = 0$ there.
- Joule–Thomson: isenthalpic expansion; $\mu_{JT} > 0$ below inversion temperature.
- Inversion temperature: $T_i = 2a/(Rb) = 2T_B$.

4. Critical Phenomena and the Continuity of State

Critical phenomena describe the behaviour of substances at the *critical point* — the unique temperature T_c , pressure P_c , and molar volume V_c at which the liquid–vapour interface vanishes and liquid and vapour become identical. Above T_c , no amount of pressure can liquefy the gas.

4.1 Phase Diagram and the Liquid–Vapour Boundary

Pressure–Temperature Phase Diagram



Key points:

- **Triple point:** all three phases coexist (unique T and P). For water: $T_{tr} = 273.16$ K, $P_{tr} = 611.7$ Pa.
- **Critical point:** liquid–vapour boundary terminates. Above T_c : single supercritical fluid (SCF) phase.
- **Vapour pressure curve:** slope given by Clausius–Clapeyron equation.

4.2 Derivation of Critical Constants from the Van der Waals Equation

Conditions at the Critical Point

At the critical point, the isotherm has an inflection point with zero slope:

$$\left(\frac{\partial P}{\partial V_m}\right)_{T_c} = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V_m^2}\right)_{T_c} = 0$$

Starting from the van der Waals EOS for 1 mol:

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

First derivative:

$$\frac{\partial P}{\partial V_m} = -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3} = 0 \quad \Rightarrow \quad RT(V_m - b)^2 = 2a \frac{(V_m - b)^2 (V_m - b)}{V_m^3} \cdot V_m^3 \quad (\text{i})$$

Second derivative:

$$\frac{\partial^2 P}{\partial V_m^2} = \frac{2RT}{(V_m - b)^3} - \frac{6a}{V_m^4} = 0 \quad \Rightarrow \quad RT(V_m - b)^3 = 3a \cdot \frac{(V_m - b)^3}{V_m^4} \cdot V_m^4 \quad (\text{ii})$$

Dividing (ii) by (i):

$$\frac{(V_m - b)}{1} = \frac{3}{2} V_m \quad \Rightarrow \quad 2V_m - 2b = 3V_m \quad \Rightarrow \quad \boxed{V_c = 3b}$$

Substituting back into (i):

$$RT_c = \frac{2a(V_c - b)^2}{V_c^3} = \frac{2a(2b)^2}{(3b)^3} = \frac{8a}{27b} \quad \Rightarrow \quad \boxed{T_c = \frac{8a}{27Rb}}$$

And using the VdW EOS at the critical point:

$$\boxed{P_c = \frac{a}{27b^2}}$$

Critical Constants (Van der Waals)

$$V_c = 3b, \quad T_c = \frac{8a}{27Rb}, \quad P_c = \frac{a}{27b^2}$$

Also useful: $\frac{P_c V_c}{RT_c} = \frac{3}{8} = 0.375$ (critical compression factor for a VdW gas).

Solved Example: Critical Constants of N₂

$a = 1.39 \text{ L}^2 \text{ atm mol}^{-2}$, $b = 0.0391 \text{ L mol}^{-1}$:

$$V_c = 3(0.0391) = 0.117 \text{ L mol}^{-1}$$

$$T_c = \frac{8(1.39)}{27(0.08206)(0.0391)} = \frac{11.12}{0.08661} = 128.4 \text{ K}$$

$$P_c = \frac{1.39}{27(0.0391)^2} = \frac{1.39}{0.04128} = 33.7 \text{ atm}$$

Experimental: $T_c = 126.2 \text{ K}$, $P_c = 33.9 \text{ atm}$ — excellent agreement.

4.3 Reduced Variables and the Law of Corresponding States

Reduced variables express T , P , and V as fractions of their critical values:

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}, \quad V_r = \frac{V_m}{V_c}$$

Law of Corresponding States (van der Waals): All gases obey the *same* equation of state when expressed in terms of reduced variables, regardless of their identity.

Substituting into the van der Waals equation:

$$\left(P_r + \frac{3}{V_r^2}\right)\left(V_r - \frac{1}{3}\right) = \frac{8}{3}T_r$$

This *reduced van der Waals equation* has no gas-specific parameters. It predicts that at the same T_r and P_r , all gases have the same Z — the basis of *generalised compressibility charts* used in engineering.

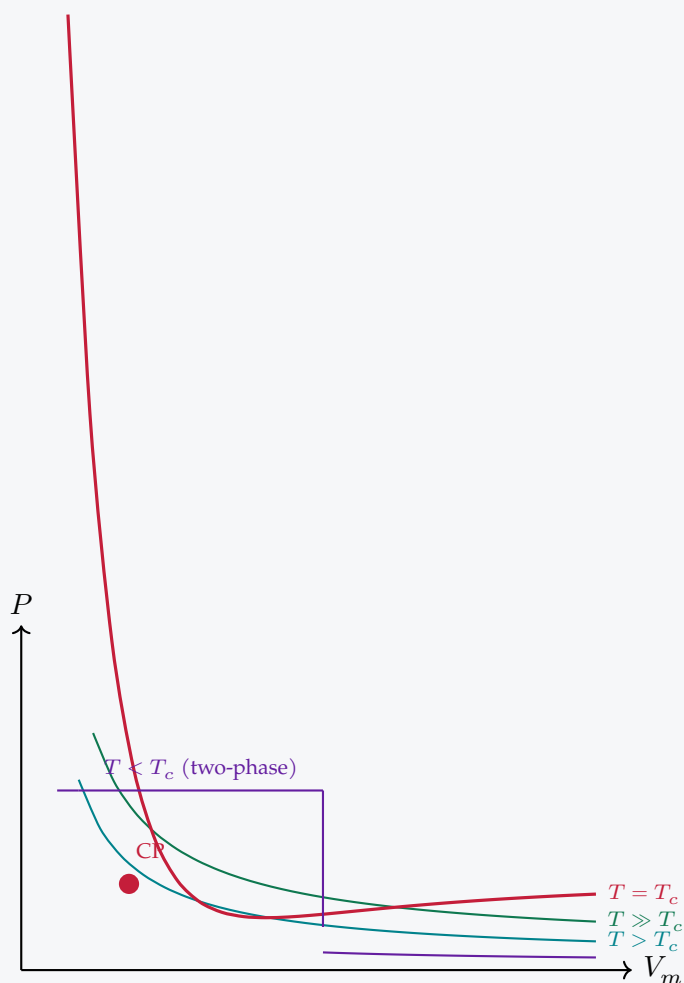
Application: Comparing States of Different Gases

CO₂ at $T = 340 \text{ K}$, $P = 100 \text{ atm}$ has $T_r = 340/304 = 1.12$ and $P_r = 100/72.9 = 1.37$.

CH₄ at $T = 215 \text{ K}$, $P = 63.5 \text{ atm}$ has $T_r = 215/191 = 1.12$ and $P_r = 63.5/46.1 = 1.38$.

By the law of corresponding states, both gases have approximately the same Z even though their absolute conditions differ.

4.4 Andrews Isotherms and the Critical Region

P - V_m Isotherms of a Real Gas (Andrews-Type)**Reading the isotherms:**

- Above T_c : smooth curves, gas cannot be liquefied.
- At T_c : inflection point at the critical point (critical isotherm).
- Below T_c : isotherms show a *horizontal portion* where liquid and vapour coexist at constant P (vapour pressure). Compression reduces volume along this plateau until all vapour condenses.

4.5 Clausius–Clapeyron Equation

Along the vapour-pressure curve, the slope dP/dT is given by the Clausius–Clapeyron equation.

Clausius–Clapeyron Equation

$$\frac{dP}{dT} = \frac{\Delta H_{\text{vap}}}{T \Delta V_{\text{vap}}} \xrightarrow{\text{ideal gas approx.}} \frac{d \ln P}{dT} = \frac{\Delta H_{\text{vap}}}{RT^2}$$

Integrated form:

$$\ln \frac{P_2}{P_1} = -\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

A plot of $\ln P$ vs $1/T$ is linear with slope $-\Delta H_{\text{vap}}/R$.

Solved Example: Vapour Pressure of Water

Water has $\Delta H_{\text{vap}} = 40.7 \text{ kJ mol}^{-1}$. At $T_1 = 373 \text{ K}$, $P_1 = 1 \text{ atm}$. Find P at $T_2 = 393 \text{ K}$.

$$\ln \frac{P_2}{1} = -\frac{40700}{8.314} \left(\frac{1}{393} - \frac{1}{373} \right) = -4893 \times (-1.364 \times 10^{-4}) = 0.668$$

$$P_2 = e^{0.668} = 1.95 \text{ atm}$$

The vapour pressure nearly doubles from 373 K to 393 K, consistent with the rapid increase seen in pressure cookers.

4.6 Supercritical Fluids

Above T_c and P_c the distinction between liquid and gas vanishes. The substance is a **supercritical fluid (SCF)** with properties intermediate between liquid and gas.

Property	Gas	SCF	Liquid
Density	Low	Liquid-like	High
Diffusivity	High	Intermediate	Low
Viscosity	Low	Low	High
Solvation	Negligible	Tunable	Fixed

Critical constants of common SCF solvents:

Solvent	T_c (K)	P_c (bar)	Applications
CO ₂	304	73.8	Decaffeination, dry cleaning, drug extraction
H ₂ O	647	220.9	Oxidation of organic waste, hydrothermal synthesis
CH ₄	191	46.1	Natural gas processing
Ethanol	516	63.8	Food industry extraction

Why sc-CO₂ is the Industrial SCF of Choice

Supercritical CO₂ has $T_c = 304 \text{ K}$ (near room temperature) and $P_c = 73.8 \text{ bar}$ (accessible with standard pumps). It is non-toxic, non-flammable, cheap, and leaves no solvent residue. Its solvating power is tunable by pressure, making it ideal for pharmaceutical extraction, polymer processing, and green chemistry.

Exam Points (Section 4)

- Derive $V_c = 3b$, $T_c = 8a/27Rb$, $P_c = a/27b^2$ from the VdW conditions at CP.
- Critical compression factor: $Z_c = P_c V_c / RT_c = 3/8$.
- Law of corresponding states: all gases follow the same reduced VdW EOS.
- Andrews isotherms: identify two-phase region, critical point, and supercritical region.
- Clausius–Clapeyron: $\ln(P_2/P_1) = -(\Delta H_{\text{vap}}/R)(1/T_2 - 1/T_1)$.
- SCF properties: liquid-like density, gas-like diffusivity; CO₂ example.

5. Applications of Intermolecular Forces and Real-Gas Behaviour

Understanding IMFs and real-gas equations of state is not purely theoretical — they govern physical properties and underpin industrial processes from gas liquefaction to drug design.

5.1 Effect of IMFs on Physical Properties

Property	How IMF controls it	Example
Boiling point	Stronger IMF $\Rightarrow$ more energy to vaporise	H ₂ O (100°C) vs H ₂ S (−60°C)
Melting point	Stronger crystal forces $\Rightarrow$ higher mp	NaCl (801°C) vs Ar (−189°C)
Viscosity	Larger/polar molecules have stronger IMF $\Rightarrow$ more resistance to flow	Glycerol $\gg$ water
Surface tension	IMF pulling surface molecules inward	Water (72 mN m ^{−1}) $\gg$ hexane (18 mN m ^{−1})
Solubility	Like-dissolves-like: matching IMF types	NaCl in water, iodine in CCl ₄
Heat of vaporisation	Energy to overcome IMF	Water 44 kJ mol ^{−1} , hexane 31 kJ mol ^{−1}

5.2 Like-Dissolves-Like Rule

Like dissolves like: A solute dissolves in a solvent when the solvent–solute interactions are comparable in type and strength to the solvent–solvent interactions. Polar solutes dissolve in polar solvents; non-polar solutes dissolve in non-polar solvents.

Thermodynamic justification: Dissolution is spontaneous when $\Delta G_{\text{solvation}} < 0$. This

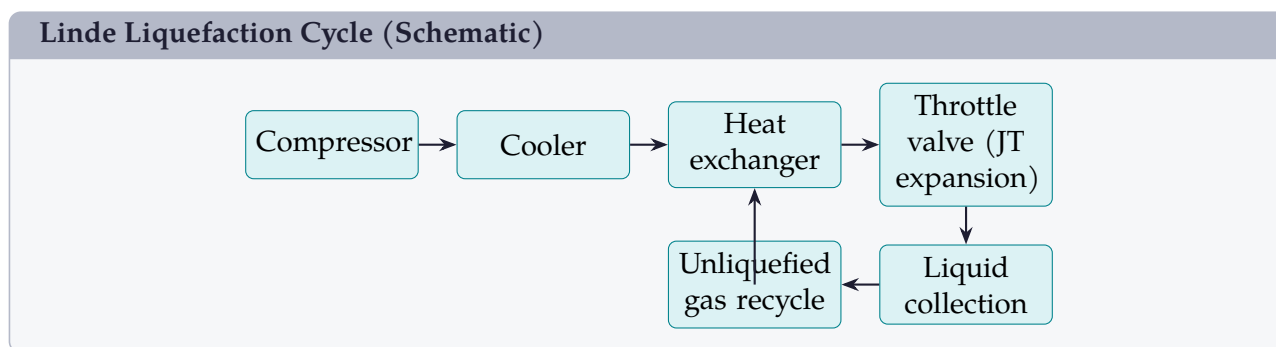
requires $\Delta H_{\text{solvation}} \approx -(\Delta H_{\text{lattice/assoc}} + \Delta H_{\text{solvent cavity}})$, which holds when IMF types match.

5.3 Hydrogen Bonding in Biological Systems

- **DNA base pairing:** A–T has 2 H-bonds (20 kJ mol^{-1}); G–C has 3 (30 kJ mol^{-1}). Collectively they stabilise the double helix.
- **Protein folding:** H-bonds between backbone N–H and C=O groups stabilise α -helices and β -sheets.
- **Enzyme catalysis:** H-bonds in the active site position substrates precisely.
- **Membrane fluidity:** Phospholipid bilayers rely on London forces between non-polar tails, H-bonds between polar heads and water.

5.4 Gas Liquefaction and Industrial Gases

Liquefying a gas requires cooling it below its critical temperature and then compressing it. The Joule–Thomson effect is harnessed industrially.



Industrial cryogenics and their T_c :

- Liquid N_2 (77 K): $T_c = 126 \text{ K}$, pre-cooling to $\sim 100 \text{ K}$ needed.
- Liquid O_2 (90 K): $T_c = 155 \text{ K}$.
- Liquid He (4.2 K): $T_c = 5.2 \text{ K}$, requires liquid N_2 pre-cool then JT expansion.
- Liquid H_2 (20 K): $T_c = 33 \text{ K}$, must pre-cool below inversion temperature.

5.5 Real Gases in Engineering Calculations

Van der Waals and virial equations are used in:

- Pipeline sizing for high-pressure natural gas ($Z \neq 1$).
- Refrigeration cycle analysis (refrigerant properties near T_c).
- Supercritical extraction unit design.
- Prediction of dew points and bubble points in process engineering.

5.6 Colligative Properties: IMF Context

Though formally under solution chemistry, colligative properties arise from the modification of solvent–solvent IMFs by added solute.

Property	IMF connection	Equation
Boiling-point elevation	Solute lowers solvent vapour pressure $\Rightarrow$ higher bp	$\Delta T_b = K_b m$
Freezing-point depression	Solute disrupts lattice IMF network	$\Delta T_f = K_f m$
Osmotic pressure	Solute concentration drives solvent flow across membrane	$\pi = iMRT$
Raoult's law	Vapour pressure lowering $\Delta P = x_{\text{sol}} P^\circ$	Based on ideal mixing of IMFs

Exam Points (Section 5)

- Boiling point order: relates directly to IMF strength (H-bond > dipole > London).
- Like-dissolves-like: IMF type determines solubility.
- DNA A-T: 2 H-bonds; G-C: 3 H-bonds — explains higher T_m of GC-rich DNA.
- Linde process: compressor $\rightarrow$ cooler $\rightarrow$ heat exchanger $\rightarrow$ throttle (JT cooling) $\rightarrow$ liquid.
- Clausius–Clapeyron gives slope of vapour-pressure curve; use to find ΔH_{vap} .

Quick Revision — Module 3

Intermolecular Forces, Real Gases and Critical Phenomena | BS-CH101

Key Formulas at a Glance

Formula	Meaning
$V_{\text{ion-ion}} = -\frac{z_1 z_2 e^2}{4\pi\epsilon_0 r}$	Coulomb energy between two ions ($\propto r^{-1}$).
$\mu = q \cdot d$	Dipole moment; 1 D = 3.336×10^{-30} C m.
$V_{\text{Keesom}} \propto -\frac{\mu^4}{k_B T r^6}$	Dipole–dipole (thermal-averaged); r^{-6} , T -dependent.
$V_{\text{London}} \propto -\frac{\alpha'_1 \alpha'_2 I_1 I_2}{r^6 (I_1 + I_2)}$	London dispersion; r^{-6} , T -independent.
$V_{\text{LJ}} = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$	Lennard-Jones 12-6 potential; $r_e = 2^{1/6}\sigma$.
$F = -dV/dr$	Force from potential energy curve.
$V_{\text{Morse}} = D_e[1 - e^{-\beta(r-r_e)}]^2$	Morse bond potential with dissociation limit.
$\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$	Van der Waals EOS (one mole).
$Z = PV_m/RT$	Compressibility factor; $Z = 1$ ideal.
$Z = 1 + B/V_m + C/V_m^2 + \dots$	Virial EOS; $B = b - a/(RT)$ for VdW.
$T_B = a/(Rb)$	Boyle temperature ($B = 0$, near-ideal).
$V_c = 3b, T_c = 8a/27Rb, P_c = a/27b^2$	Critical constants (VdW).
$Z_c = P_c V_c / RT_c = 3/8$	Critical compressibility factor (VdW).
$T_r = T/T_c, P_r = P/P_c, V_r = V_m/V_c$	Reduced variables.
$(P_r + 3/V_r^2)(V_r - 1/3) = 8T_r/3$	Reduced VdW EOS (universal).
$\mu_{\text{JT}} = (\partial T / \partial P)_H$	Joule–Thomson coefficient.
$T_i = 2a/(Rb) = 2T_B$	Inversion temperature.
$\ln(P_2/P_1) = -(\Delta H_{\text{vap}}/R)(1/T_2 - 1/T_1)$	Clausius–Clapeyron (integrated).
$\Delta T_b = K_b m; \Delta T_f = K_f m; \pi = iMRT$	Colligative properties.

One-Line Definitions

- **Intermolecular force:** Attractive or repulsive force between separate molecules/ions.
- **Dipole moment:** Vector quantity $\mu = qd$; non-zero for asymmetric charge distribution.
- **London dispersion:** IMF due to instantaneous and induced temporary dipoles; exists in all molecules.
- **Keesom interaction:** Thermal-averaged attraction between two permanent dipoles; proportional to r^{-6} and $1/T$.
- **Debye interaction:** Attraction of permanent dipole on an induced dipole; r^{-6} , temperature-independent.
- **Hydrogen bond:** Strong directional interaction of H (bonded to F/O/N) with a lone pair; 15–40 kJ mol⁻¹.
- **Polarisability:** Ease of distortion of an electron cloud by an electric field; increases down a group.
- **Lennard-Jones potential:** Empirical 12-6 pair potential combining repulsion and London dispersion.
- **Equilibrium separation (r_e):** Distance at which V is minimum and $F = 0$.
- **Potential energy surface:** Multi-dimensional energy landscape as function of molecular geometry.
- **Transition state:** Saddle point on the PES; highest energy along the reaction path.
- **Activation energy (E_a):** Energy barrier separating reactants from transition state.
- **Real gas:** Gas deviating from ideal due to finite molecular volume and attractions.
- **Van der Waals a :** Correction for intermolecular attractions (reduces pressure).
- **Van der Waals b :** Excluded volume per mole due to finite molecular size (reduces free volume).
- **Compressibility factor (Z):** PV_m/RT ; equals 1 for ideal gas.
- **Virial coefficient B :** Second virial coefficient; zero at Boyle temperature.
- **Boyle temperature:** $T_B = a/(Rb)$; gas behaves nearly ideally.
- **Critical temperature (T_c):** Highest T at which gas can be liquefied by pressure alone.
- **Critical pressure (P_c):** Pressure needed to liquefy gas at T_c .
- **Critical point:** State where liquid and vapour become identical; (T_c, P_c, V_c) .
- **Reduced variables:** T_r, P_r, V_r ; enable law of corresponding states.
- **Law of corresponding states:** All gases obey the same reduced EOS.
- **Joule–Thomson effect:** Temperature change of real gas on isenthalpic expansion through a throttle.
- **Inversion temperature (T_i):** $\mu_{JT} = 0$; $T_i = 2a/(Rb)$.

- **Supercritical fluid:** Phase above T_c and P_c with liquid-like density and gas-like diffusivity.
- **Clausius–Clapeyron equation:** Relates vapour pressure change to temperature via ΔH_{vap} .
- **Like-dissolves-like:** Solubility is maximised when solute–solvent IMF type matches.
- **Linde process:** Industrial gas liquefaction using Joule–Thomson expansion cycle.

Frequently Asked / Viva Questions

Q1. What are intermolecular forces? Arrange them in increasing strength.

Forces between molecules/ions: London < dipole–dipole < H-bond < ion–dipole < ion–ion.

Q2. Why do London forces increase down a noble-gas group?

Larger atoms have more electrons and higher polarisability; instantaneous dipoles are stronger.

Q3. Why is CO₂ non-polar despite having polar bonds?

Linear geometry makes bond-dipole vectors cancel exactly.

Q4. What is Keesom interaction and why is it temperature-dependent?

Dipole–dipole orientation averaged over thermal motion; higher T scrambles alignment, weakening the interaction.

Q5. State the Lennard-Jones potential and identify each term.

$V_{\text{LJ}} = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$; first term: repulsion; second: London attraction.

Q6. What is the equilibrium separation in the LJ potential?

$r_e = 2^{1/6}\sigma \approx 1.122\sigma$; found by setting $dV/dr = 0$.

Q7. What is a potential energy surface?

A multi-dimensional surface showing molecular energy as a function of geometry; minima = stable species, saddle points = transition states.

Q8. Write the van der Waals equation and explain a and b .

$(P + a/V_m^2)(V_m - b) = RT$; a corrects for attractions (adds to P); b corrects for finite molecular volume (subtracts from V).

Q9. What does a compressibility factor $Z < 1$ indicate?

Attractive forces dominate; gas occupies less volume than ideal.

Q10. State the virial EOS.

$Z = 1 + B/V_m + C/V_m^2 + \dots$; B is zero at the Boyle temperature.

Q11. Derive the Boyle temperature from the VdW EOS.

At T_B , second virial coefficient $B = b - a/(RT_B) = 0$, giving $T_B = a/(Rb)$.

Frequently Asked / Viva Questions (contd.)

Q12. Derive the critical constants from the VdW EOS.

Set $(\partial P/\partial V_m)_T = 0$ and $(\partial^2 P/\partial V_m^2)_T = 0$; solving gives $V_c = 3b$, $T_c = 8a/27Rb$, $P_c = a/27b^2$.

Q13. What is the critical compressibility factor for a VdW gas?

$$Z_c = P_c V_c / RT_c = 3/8 = 0.375.$$

Q14. State the law of corresponding states.

All gases obey the same reduced EOS $(P_r + 3/V_r^2)(V_r - 1/3) = 8T_r/3$ when variables are expressed relative to critical values.

Q15. What is the Joule–Thomson effect?

Isenthalpic ($\Delta H = 0$) expansion of a real gas through a throttle; temperature falls below the inversion temperature, rises above it.

Q16. What is the inversion temperature?

$$T_i = 2a/(Rb) = 2T_B; \text{ above } T_i \text{ the JT coefficient is negative (gas warms).}$$

Q17. Why must hydrogen be pre-cooled before JT liquefaction?

$T_i(\text{H}_2) \approx 200 \text{ K}$, below room temperature; at room temperature $\mu_{JT} < 0$, so H_2 warms on throttling.

Q18. State and explain the Clausius–Clapeyron equation.

$d \ln P/dT = \Delta H_{\text{vap}}/(RT^2)$; relates slope of vapour-pressure curve to enthalpy of vaporisation; $\ln P$ vs $1/T$ is linear.

Q19. What is a supercritical fluid? Give one industrial example.

Fluid above T_c and P_c with tuneable density; supercritical CO_2 ($T_c = 304 \text{ K}$, $P_c = 73.8 \text{ bar}$) used for caffeine extraction.

Q20. How does hydrogen bonding stabilise the DNA double helix?

A–T: 2 H-bonds; G–C: 3 H-bonds; collectively millions of H-bonds along the helix provide thermodynamic stability while remaining individually reversible for replication.

Q21. Why is water anomalous among hydrides?

Extensive H-bonding raises bp (100°C vs -60°C for H_2S), increases heat of vaporisation, and makes ice less dense than liquid water.

Q22. What is the Andrews isotherm?

P – V_m curve of a real gas at various temperatures; shows two-phase plateau below T_c and inflection point at T_c .

Q23. Define triple point and critical point.

Triple point: coexistence of solid, liquid, and gas at unique (T_{tr}, P_{tr}) . Critical point: end of liquid–vapour boundary at (T_c, P_c, V_c) .

Summary Reference Table — Module 3

Topic	Key concept	Key formula	Exam output
IMF types	6 types from ionic to London	$V \propto r^{-1}$ to r^{-6}	Strength order + examples
Lennard-Jones	Repulsion + dispersion	$4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$	Diagram + labels
PES	Energy vs geometry	Saddle point = TS	Reaction-path diagram
Van der Waals	Real-gas corrections	$(P + a/V_m^2)(V_m - b) = RT$	Worked numerical
Compressibility	Deviation from ideal	$Z = PV_m/RT$	Z vs P sketch
Boyle temperature	$Z \approx 1, B = 0$	$T_B = a/(Rb)$	Calculation
Critical constants	From VdW conditions	$V_c = 3b, T_c = 8a/27Rb$	Derivation or value
Corresponding states	Universal VdW	$(P_r + 3/V_r^2)(V_r - 1/3) = 8T_r/3$	State the law
Joule–Thomson	Isenthalpic expansion	$\mu_{JT} = (\partial T/\partial P)_H$	Table: T vs T_i
Clausius–Clapeyron	Vapour-pressure slope	$\ln(P_2/P_1) = -\frac{\Delta H}{R}\Delta(1/T)$	Find P or ΔH
Supercritical fluid	Above T_c, P_c	$T_c(\text{CO}_2) = 304 \text{ K}$	Applications